

HW #1: Mixture Density Networks

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Probabilistic Deep Learning — Fall 2026

1 Model and implementation

For an input $\mathbf{x}$ and target $\mathbf{y}$, our mixture density network (MDN) models the conditional density as

$$p(\mathbf{y} | \mathbf{x}) = \sum_{k=1}^K \pi_k(\mathbf{x}) \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_k(\mathbf{x}), \boldsymbol{\Sigma}_k(\mathbf{x})), \quad \sum_{k=1}^K \pi_k(\mathbf{x}) = 1. \quad (1)$$

The network is trained by minimizing the mean negative log likelihood (NLL). To implement this, we created an NN module that outputs the mixture weights (through a softmax), μ , and Σ (through a softplus, plus an additional term to ensure a minimum scale of $\epsilon > 0$) for each mixture component. We optimize using the log-sum-exp expansion. Our hyperparameters include the number of mixture components K (the number of Gaussians in the mixture), the rank of the covariance matrices (4), learning rate (1e-3), batch size (64), and number of epochs (2000).

2 Synthetic two-dimensional example

The data we used for our 2D synthetic example was generated by sampling 4000 points from two different Gaussian distributions with different means and covariances. Then, we added Gaussian noise to the output. This resulted in a multimodal distribution for the output given the input, see Figure 1. This resulting trained model should show a single Gaussian near the intersection of the distributions, and two separate Gaussians that are increasingly far apart as the input moves away from the intersection (if the MDN learns well). Figure 2 does indeed show this behavior. The MDN adequately finds each mode with respect to the data.

Additionally, we tried a range of values for K , see Figure 3, and found that there was no significant improvement for any K greater than 2. This is expected because the data comes from a simple two-modal distribution.

We also compare these K models to a deterministic regressor. This regressor performed similarly to the $K = 1$ MDN. NLL and RMSE are reported for all the aforementioned models in Table 1, when relevant. The test RMSE is lowest for $K = 5$, at 0.9054, but $K = 2$ is very close at 0.9056.

We observe that training loss (NLL or MSE) over epochs decreases for all models, and that MDN models with $K > 1$ outperform the deterministic and single-Gaussian models. Models with $K \in 2, 3, 4, 5$ perform similarly, with no clear best choice.

Table 1: Synthetic-data performance loaded directly from `csvs/synthetic_summary.csv`. NLL is not defined for the deterministic model.

Model	K	Train NLL	Test NLL	Train RMSE	Test RMSE
Deterministic	—	N/A	N/A	0.914328	0.920263
Single Gaussian	1	0.832287	0.807886	0.921331	0.905627
MDN	2	-0.257680	-0.263095	0.920119	0.907287
MDN	3	-0.263673	-0.265676	0.919707	0.909541
MDN	4	-0.262717	-0.265303	0.919435	0.910978
MDN	5	-0.262428	-0.257620	0.920986	0.905491

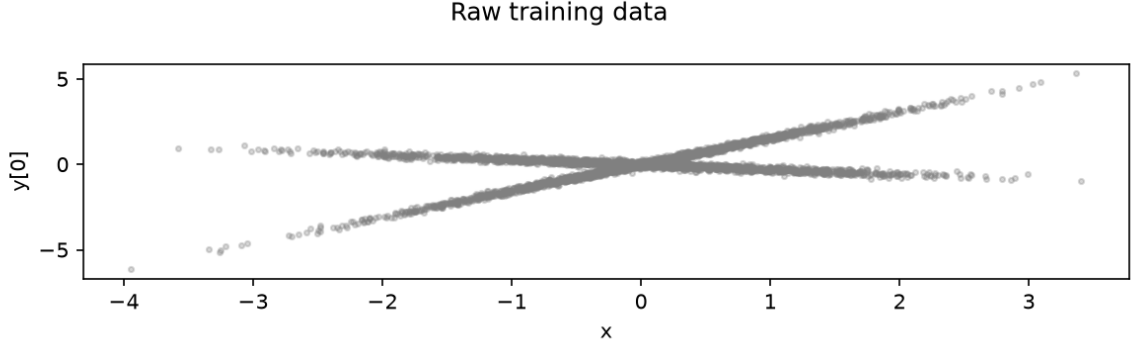


Figure 1: Raw training observations for the synthetic task, showing the multiple output modes associated with the input.

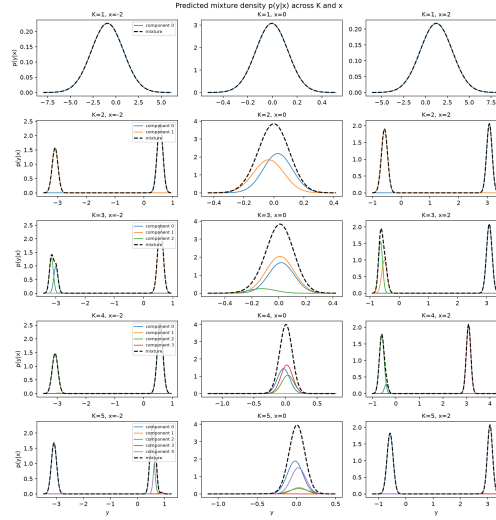


Figure 2: Learned conditional density for the synthetic 2D task over a grid of input and output values. The density should reveal the modes captured by the MDN and how their locations, weights, and widths vary with the input.

3 Higher-dimensional realistic data

We use the Wind Turbine SCADA dataset available on Kaggle as the realistic data set. This data set contains the features described in Table 2. To construct a trainable feature set we decided to drop the Theoretical Power Curve, as we are unsure how the manufacturer generated this feature. We also replaced the raw Date/Time timestamp with cyclical (sine/cosine) encodings of hour-of-day and day-of-year, mapping each periodic time feature onto the unit circle via $\theta = 2\pi t/\text{period}$ (period = 24 for hour, using the fractional hour to account for the dataset’s 10-minute sampling; period = 365 for day-of-year) so that adjacent times (e.g. 24:00 and 00:01) remain close in feature space rather than being separated by the full width of the raw numeric scale. Wind speed and wind direction were kept as direct scalar inputs.

Table 2: Features of the wind turbine SCADA dataset.

Feature	Unit	Description
Date/Time	–	Timestamp, recorded at 10-minute intervals
LV ActivePower	kW	Power generated by the turbine at that moment
Wind Speed	m/s	Wind speed at the hub height of the turbine
Theoretical Power Curve	kWh	Theoretical power the turbine generates at a given wind speed, as specified by the manufacturer
Wind Direction	°	Wind direction at the hub height of the turbine

Figure 4 shows each feature plotted against the target, LV ActivePower. From a visual perspective, most features did

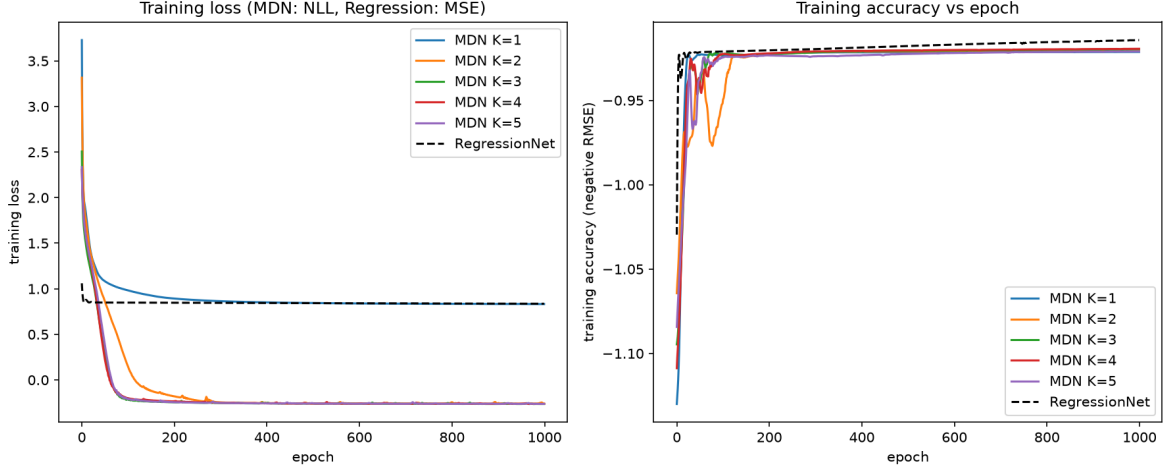


Figure 3: Training objective and RMSE across epochs for the deterministic baseline, single-Gaussian model, and MDNs with different numbers of components.

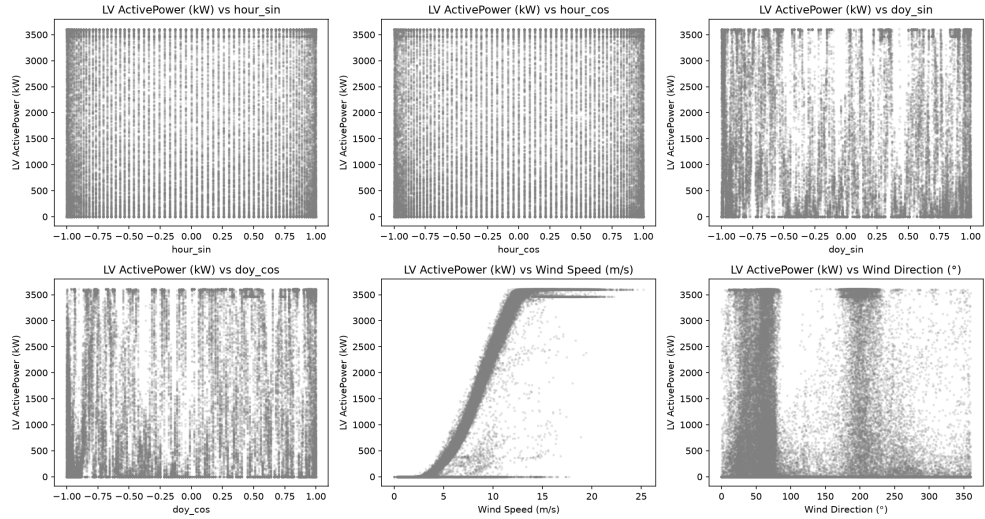


Figure 4: Wind-turbine active power against each available numeric predictor, used to assess multimodality and heteroscedasticity before fitting the models.

not obviously display a multimodal relationship with respect to ActivePower except for Wind Speed. In most cases the entire range of ActivePower can be observed for nearly all feature values, however, Wind Speed does display a multimodal relationship with the target: across the ramp region (roughly 4-13 m/s) and at or near maximum power output (13 m/s and beyond). Wind Direction shows a related but weaker pattern, power appears to switch between near-zero and near-max output across direction bands, in particular straddling 200° , but this on/off behavior is not by itself evidence of a genuinely multimodal conditional distribution. Figure 5 shows the conditional distribution of ActivePower within narrow bands straddling several representative query points: 5 ± 0.5 , 9 ± 0.5 , and 13 ± 0.5 m/s for Wind Speed, $50 \pm 10^\circ$ and $200 \pm 10^\circ$ for Wind Direction. Unlike the scatter in Figure 4, this makes any separation in $P(\text{ActivePower} | x_0)$ explicit as distinct histogram peaks rather than something inferred from the shape of a scatter cloud.

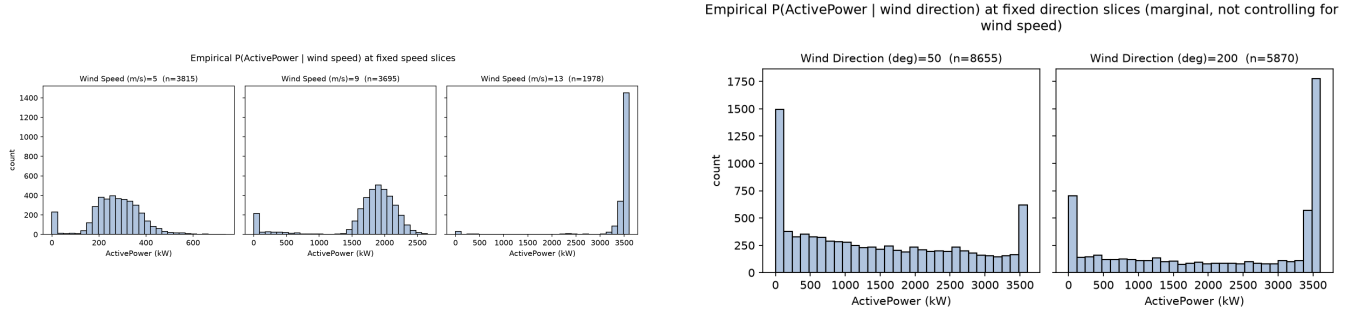


Figure 5: Conditional histograms of ActivePower at fixed slices of wind speed (left) and wind direction (right).

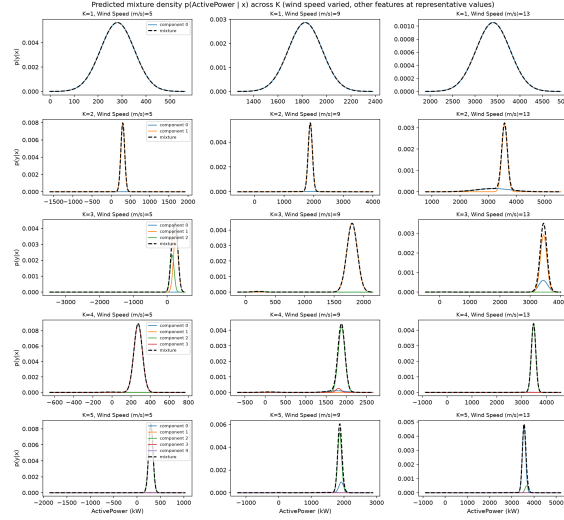


Figure 6: Learned conditional density for the wind-turbine task over a grid of inputs and active-power outputs. Use this view to show heteroscedasticity, multiple operating regimes, and regions where the model assigns diffuse or multimodal probability.

Compare training and test behavior on the realistic dataset. Discuss generalization gaps, predictive sharpness and calibration, regions of high uncertainty, the effect of K , and how conclusions differ from the controlled synthetic example. Reference Figures 4 and 6, Figure 7, and Table 3.

Table 3: Wind-turbine performance loaded directly from `csvs/wind_turbine_summary.csv`. RMSE is reported in kW; NLL is evaluated on standardized targets and is not defined for the deterministic model.

Model	K	Train NLL	Test NLL	Train RMSE	Test RMSE
Deterministic	—	N/A	N/A	321.065277	324.760101
Single Gaussian	1	-0.792758	-0.708927	349.651184	353.919952
MDN	2	-1.085540	-1.080853	354.051941	358.169586
MDN	3	-1.039635	-1.022697	358.404694	361.552246
MDN	4	-1.377420	-1.371821	351.816528	356.182159
MDN	5	-1.437616	-1.436583	348.602081	353.592316

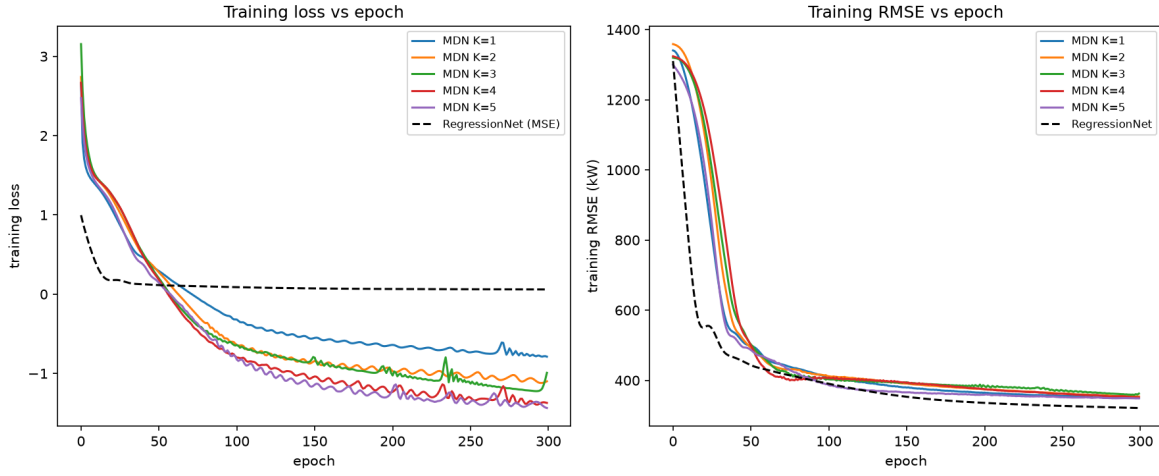


Figure 7: Training and validation/test behavior for the realistic-data models, including the selected checkpoint or stopping point.

4 Conclusion

Summarize when the MDN improves on deterministic and single-Gaussian regression, identify the best-supported mixture count, state limitations, and give one concrete next experiment.